Yuchen Wang

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Education

University of Illinois Urbana-Champaign, Computer Science — Champaign, IL, USA, 2025 – 2027

Peking University, Intelligent Science and Technology (Artificial Intelligence) — Beijing, China, 2021 – 2025
- Zhi Class (Elite Program), Top 10%

Experience

Alibaba U.S., Research Intern — LLM Training Algorithms & Infrastructure — Sunnyvale, CA, USA, May 2026 – Dec 2026, 8 months
- Core contributor to Occamy-1.0 (35B-A3B); co-designed staged SFT/HDPO, model soup, and SAO with trainable context compaction and cross-segment credit assignment. 500K+ downloads across Occamy variants.
- Architected environment-/capability-first data synthesis and multi-harness training infrastructure with state-aware graders and asynchronous actor–critic learning; the deduplicated SFT union spans 15K trajectories and 403M tokens.
- Shipped Occamy with Accio; versus its Qwen3.6 base, the released model raised AutomationBench strict pass from 7.5% to 27.6% and Claw-Eval average from 69.5 to 82.2, while reducing Claw-Eval T/C full-trial time by 36.6%.

Freedo Technology, Research Intern — 3D Vision and Generative AI — Beijing, China, May 2025 – Aug 2025, 4 months
- Built an end-to-end depth-conditioned ControlNet pipeline for 3D reconstruction from noisy point clouds, improving geometric fidelity from 32.7% to 85.3%.

Open Source Projects

Open-Source LLM Systems Engineering

RL infrastructure, long-context training, inference optimization, optimizer correctness, and training-data pipelines across NVIDIA NeMo, Megatron-LM, vLLM, SGLang, FlashInfer, and Hugging Face.
- RL infrastructure: bypassed the Ray driver for NeMo RL teacher top-k transfers, achieving 4.44–5.31× speedup in controlled 8×B200 transfer measurements. Corrected async GRPO/distillation checkpoint-resume bookkeeping in TRL. Focused CPU checks passed, with end-to-end training-resume validation pending.
- Long-context training: developing Qwen4-Exp integration in Megatron Bridge for QSA, per-layer embeddings, gated residuals, and HF conversion; GPU integration remains pending. Added fused GDN Q/K normalization and selective Mamba recompute in Megatron-LM. Made packed QSA selection document-local in mcore-bridge, 2.70× faster on an 8K synthetic selector case.
- Inference memory and correctness: in-place BF16 MoE conversion in vLLM halved 120B TP2 conversion peak allocation from 7.88 to 3.94 GiB; fixed calibrated FP8 KV prefill in FlashInfer. Fixed Mamba cache-publication ordering in SGLang.
- Optimizer correctness: fixed small-norm normalization in Muon and implemented checkpoint-safe Adam counters in Megatron-LM. CPU/DCP round-trip checks passed; full CUDA training validation remains pending.
- Training data: preserved source sharding in HF Datasets streaming shuffle, restoring all four DataLoader workers on 19 source shards, with deterministic epoch and checkpoint-restoration coverage.

Projects

Dynamic Prefill Optimization via Adaptive Online Packing
AIMD control with live p95 TTFT feedback and greedy/DP prompt packing.
- Achieved up to 20% lower TTFT on production-style DynamoLLM traces.

FlashAttention-style CUDA Optimization for GPT-2
Tiled online softmax, memory-efficient attention, and kernel fusion.
- Reduced HBM traffic by roughly 10× and achieved up to 9% end-to-end speedup.

Reinforcement Learning for Legal Reasoning
Advised by Prof. Yansong Feng at Peking University; Zero-RL → distilled-CoT SFT → GRPO for structured legal multiple-choice reasoning.
- Reached 57.6% accuracy, outperforming SFT-only and RL-only baselines.

Academic Service

- Reviewer, WSDM 2027.

Awards

Merit Student

Zhi Class Scholarship

CMO (Provincial First Prize), 2020

Skills

- Python, C++, CUDA, PyTorch, Ray, FSDP, Megatron-LM, NeMo RL, vLLM, SGLang, Nsight, RL, distributed training, diffusion transformers, ControlNet, 3D vision, and point clouds